

Buckingham Pi Theorem

Definition 1 (Buckingham Pi Theorem). Any physical relation involving N dimensional variables can be written in terms of a complete set of $N - r$ independent dimensionless variables, where r is the rank of the dimensional matrix $\mathcal{D}$.

The notational convention for the Buckingham Pi Theorem is that the “pi’s”, $\Pi_1, \dots, \Pi_{N-r}$ represent dimensionless variables and a relation between them is given by $F(\Pi_1, \dots, \Pi_{N-r}) = 0$.

Problems

T₇¹ Last Thanksgiving I called my mom to ask how long I should roast my 2 kg turkey in the oven. She said 2 hours. Several extra guests were added and I ended up cooking a turkey with mass 8 times larger, 16 kg. Rather than making another call to my mom, we will use **dimensional analysis** to determine how long I should have cooked the turkey.

The heat equation is

$$\frac{\partial T}{\partial \tau} = -\frac{k}{c\rho} \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right],$$

where

- k = thermal conductivity
- c = specific heat

Because when we change the turkey to a larger one, the mass density doesn’t change, we need an explicit expression of the mass into the equation:

- $\rho = m/V$ mass density
- m is the mass of the turkey
- V is the volume of the turkey

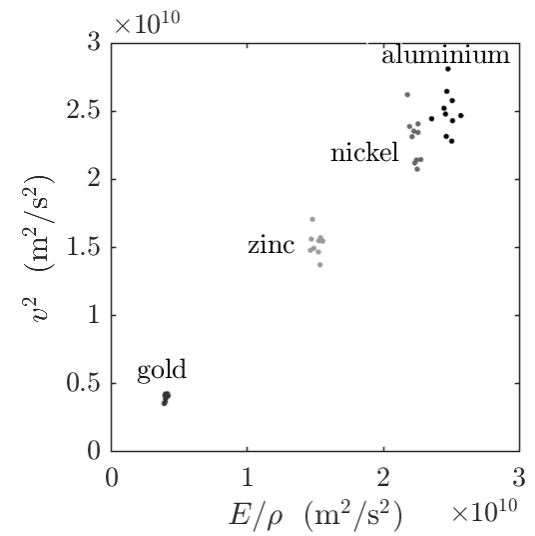
- Observe that k and c don’t relate to our objective: cooking time vs turkey size. So we can define $\beta = \frac{k}{c}$ as a variable. Use the PDE to find its units.
- What are the relevant variables we need to consider and what are their units?
- What is the dimensional matrix $\mathcal{D}$, its rank and nullity?
- Find a bases for its null space.
- Use the Buckingham Pi Theorem to conclude a relation between the variables.
- Use this relation to find the cooking time of the larger turkey.

Hint: When you consider the larger turkey, remember that the mass density remains constant. What does this imply?

¹Based on the 2019 test 1.

T₈² A compressive wave propagates with speed v (a length per time) through a long solid metal bar. The bar has a density ρ (a mass per volume) and a compressive stiffness, known as the Young's modulus, E (a pressure or a force per area).

- Use the Buckingham Pi Theorem to determine a relationship among these three variables.
- Use the measurements of the wave speed in various metal bars plotted below to determine any unknown functions.



(Challenge problem) Find how the speed of a surface wave in deep ocean water is related to its wavelength using dimension analysis. How does doubling the wavelength change the speed of the wave?

Hint: You have to consider good variables for it to work.

²From the 2022 test 1.